



ENSAI CINEMA CLUB

Analysis File

Étudiants :

N. Bougis,
S. Mironescu
Z. Goulamaly
S. Harris

Tuteur :
C. Valot

Summary

Introduction	2
Part 1 – Functional Analysis	2
Rephrasing the requirements	2
Selected features	2
Priority features	2
Features included if time allows	2
Features excluded	2
Summary : Use case diagram	2
Part 2 – Technical-Functional Layout	2
Benchmark of the Movie Information API	2
Context	2
Comparison of the Three APIs	2
Exclusion of IMDb	2
JSON Comparison — OMDb vs TMDB	3
Example JSON Payloads	3
OMDb	3
TMDB (language=fr-FR)	3
Final Decision	4
Overall structure and integration with third-party services	5
Justification of technical choices	5
Part 3 – Technical Design	5
Architecture diagram	5
Data model	5
Class diagram	5
Activity diagrams	5
Part 4 – Project management description	6
Team organization	6
Task assignment	6
Methodology	7
Gantt diagrams	7
Phase 1 : Analysis (28/08 - 17/09)	7
Phase 2 : Development (17/09 - 21/11)	8
Challenges encountered and adjustments	8
Conclusion	8

Introduction

Part 1 – Functional Analysis

Rephrasing the requirements

Selected features

Priority features

Features included if time allows

Features excluded

Summary : Use case diagram

Part 2 – Technical-Functional Layout

Benchmark of the Movie Information API

Context

Three movie data APIs are under consideration for the project : **IMDb**, **OMDb**, and **TMDB**. The goal is to compare them, rule out any unsuitable options, and select the one that provides the most relevant information based on the transmitted JSON (e.g., searching for information on the movie *The Odyssey*).

Comparison of the Three APIs

Criteria	OMDb	TMDB	IMDb
Price	Free (1000 req./day)	Free (40 req./10s) or paid upon negotiation	Paid (AWS, min. \$150,000)
Official Public API	Yes	Yes	No
Ease of Use	Easy	Easy	Difficult (AWS)
Data Completeness	Complete	Very complete	Very complete
License	Commercial use allowed	Non-commercial use by default	Mandatory commercial license
Language	English only	French supported (fr-FR)	French supported

Exclusion of IMDb

IMDb is ruled out for the following reasons :

- No official public API available : access is provided via AWS with a minimum entry cost of **\$150,000**.
- Incompatible with the project's budget and technical requirements.

The evaluation therefore focuses solely on **OMDb** and **TMDB**.

JSON Comparison — OMDb vs TMDB

Comparison based on the same search query (*"The Odyssey"*), showing the relevant data returned by each API :

Data Field	OMDb (JSON)	TMDB (JSON, language=fr-FR)
Title	"Title": "The Odyssey" (English only)	"title": "L'Odyssée" (translated into French)
Synopsis	"Plot" in English only	"overview" translated into French
Genres	"Genre": "Action, Adventure, Fantasy" (English text)	"genres": [{"id_genre": "Aventure"}, ...] (translated)
Runtime	"Runtime": "172 min"	"runtime": 172 (minutes)
Release Date	"Released": "17 Jul 2026"	"release_date": "2026-07-15"
Content Rating	"Rated": "R" : US age rating classification	"adult": false : indicates if the movie is rated adult/pornographic "softcore": false : specific flag designating non-explicit erotic content

Example JSON Payloads

OMDb

```
1 {  
2   "Title": "The Odyssey",  
3   "Plot": "A retelling of Homer's epic tale of Odysseus's journey home.",  
4   "Genre": "Action, Adventure, Fantasy",  
5   "Runtime": "172 min",  
6   "Released": "17 Jul 2026",  
7   "Rated": "R"  
8 }
```

TMDB (language=fr-FR)

```
1 {  
2   "title": "L'Odyssée",  
3   "overview": "Le retour périlleux d'Ulysse après la guerre de Troie.",  
4   "genres": [  
5     { "id_genre": "Aventure" },  
6     { "id_genre": "Action" },  
7     { "id_genre": "Fantastique" }  
]
```

```
8 ],
9 "runtime": 172,
10 "release_date": "2026-07-15",
11 "adult": false,
12 "softcore": false
13 }
```

Final Decision

TMDB is selected as the reference API for the project.

It is the only one of the two candidates to return key fields — title, synopsis, genres — **translated into French** via the `language=fr-FR` parameter, whereas OMDb strictly provides these fields in English.

Overall structure and integration with third-party services

Justification of technical choices

Part 3 – Technical Design

Architecture diagram

Data model

Class diagram

Activity diagrams

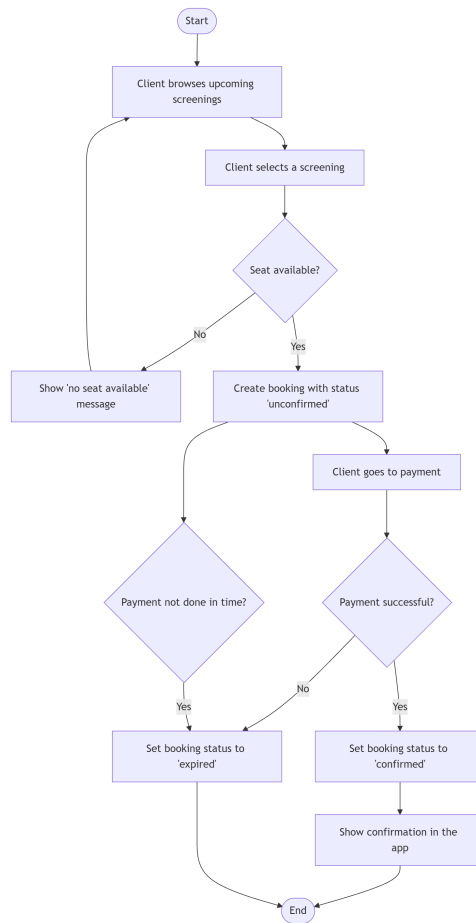


FIGURE 1 : Activity diagram : a client booking a screening

This diagram shows the booking process from the client's side : picking a screening, checking seat availability, paying, and getting a confirmation or an expired booking if the payment fails or takes too long.

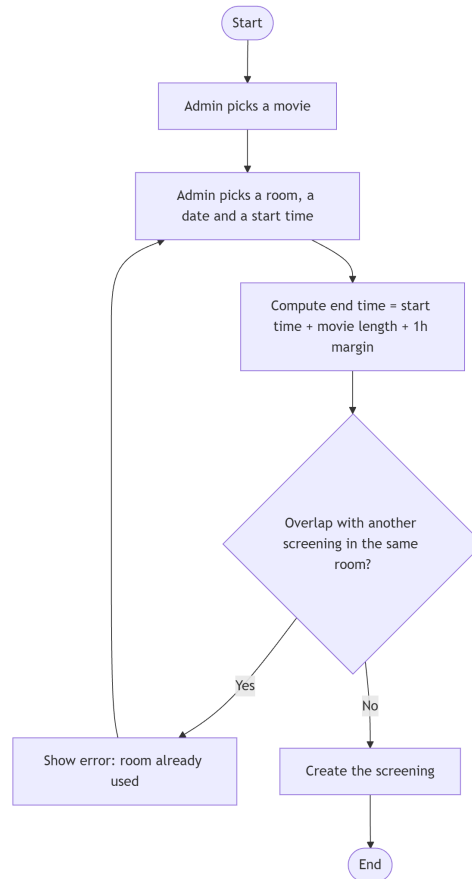


FIGURE 2 : Activity diagram : an admin scheduling a screening

This diagram shows how an admin adds a new screening to the planning. The main step is checking that the room is free, with at least 1 hour of margin before and after other screenings.

Part 4 – Project management description

Team organization

We are 4 people in the team and each one has a role for the whole project :

Role	Person
Project Manager	Nolann
Technical Director	Simon
DAO Supervisor	Samuel
Testing Supervisor	Zoeb

Task assignment

We split the work mostly along these roles. Nolann handles coordination, planning and keeping the Gantt diagram up to date. Simon takes care of the technical side, meaning the architecture, the class diagram and the general design choices. Samuel is more on the data side, so the database and the data model, and how we access the data. And Zoeb is in charge of testing, basically making sure stuff actually works before we call it done.

Of course this is not a strict separation : everyone can work on any part if needed, especially since we're only 4 people. But these roles give each of us a main area to focus on and to be responsible for.

Methodology

We have a team meeting every week to check where everyone is, discuss blocking points and decide what to do next. For now we mostly talk on WhatsApp when we need to decide something quickly, and we put our documents on Google Drive so everyone can access them. We know that WhatsApp is not the best communication tool for this sort of project work , so we're currently looking for a possibly more efficient tool, but we haven't switched yet.

The code is on GitHub (github.com/nolannb35/ENSAI-2A-projet-info-team6), we use it to keep track of the different parts of the project and work at the same time without breaking each other's code.

We also want to do some kind of review before the big deadlines (Analysis File, immersion, final delivery), just to check everyone's part actually fits together before it's too late to fix.

Gantt diagrams

Phase 1 : Analysis (28/08 - 17/09)

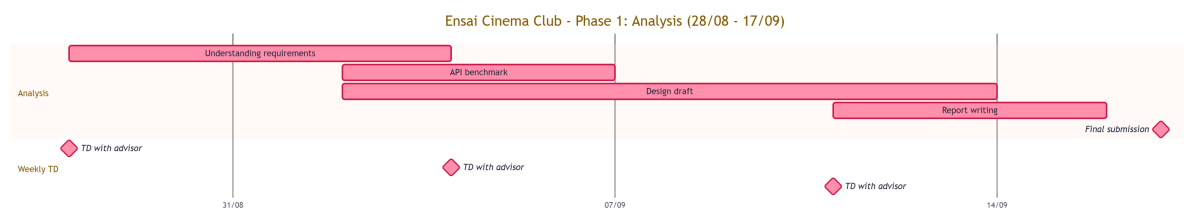


FIGURE 3 : Gantt chart for the analysis phase (28/08 - 17/09)

This phase goes from the start of the project to the Analysis File deadline. We start by understanding the subject and turning it into real features, then we do the movie API benchmark at the same time as a first draft of the technical design, and we finish by writing and reviewing the report. The three TD meetings are basically checkpoints with our advisor, where we can still adjust the plan if something's off.

Phase 2 : Development (17/09 - 21/11)

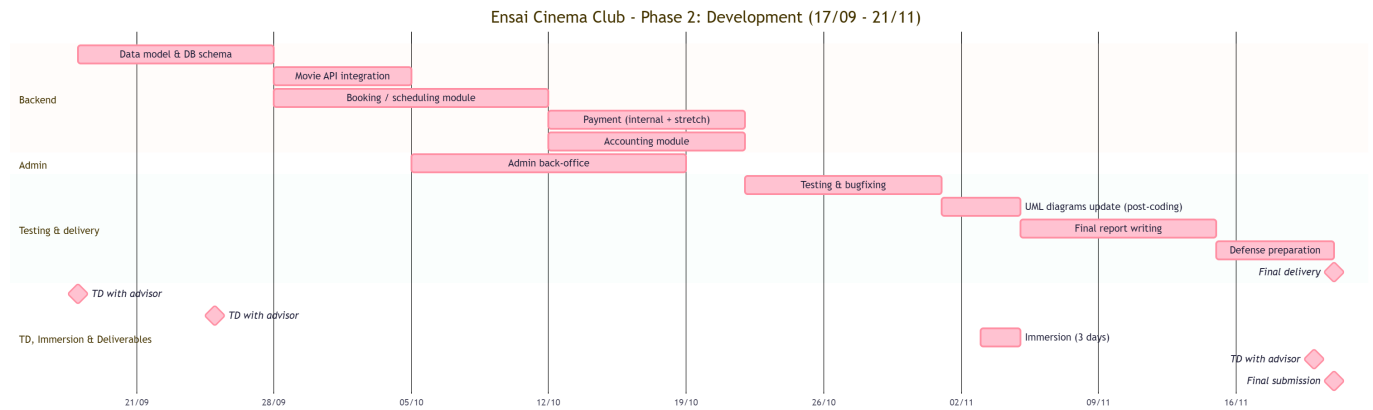


FIGURE 4 : Gantt chart for the development phase (17/09 - 21/11)

This phase is where we actually code stuff. We start with the backend (database, movie API, booking, payment, accounting), then the admin back-office, and we finish with testing, updating the UML diagrams to match what we actually coded, writing the final report and prepping the defense. The 3-day immersion in early November is when we're expecting to move a lot faster than usual since we'll have way more time on it. That's basically it, everything ends up pointing to the final submission on 21/11.

Challenges encountered and adjustments

Conclusion